



ALLIANCE
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NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Assignment No.: 1

Importing Business Datasets from Excel/CSV & MySQL into Power BI

Problem Statement

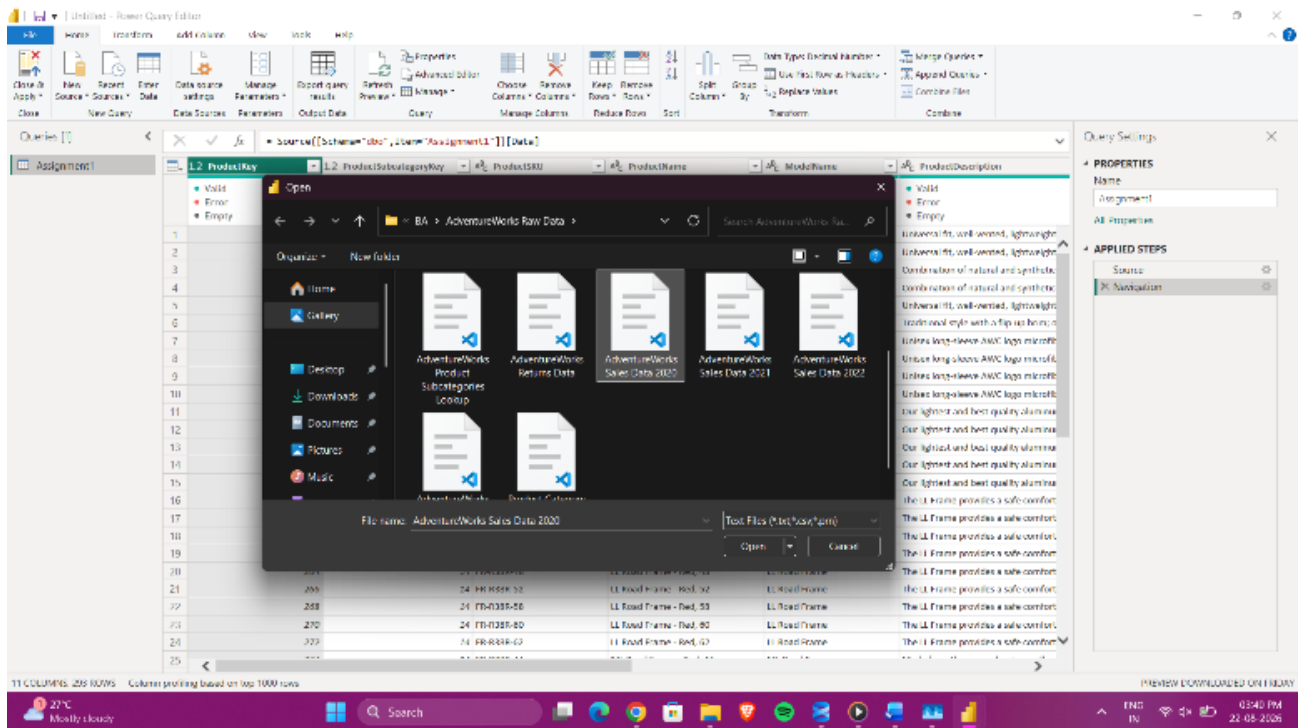
A retail company stores its business data across flat files (Excel/CSV) and a MySQL database. As a data analyst, connect Power BI Desktop to these data sources, import the required data, explore the Power BI interface, and create basic visuals to analyze sales by product, store, customer, and time.

Methodology / Steps Followed

Step 1: Data Collection

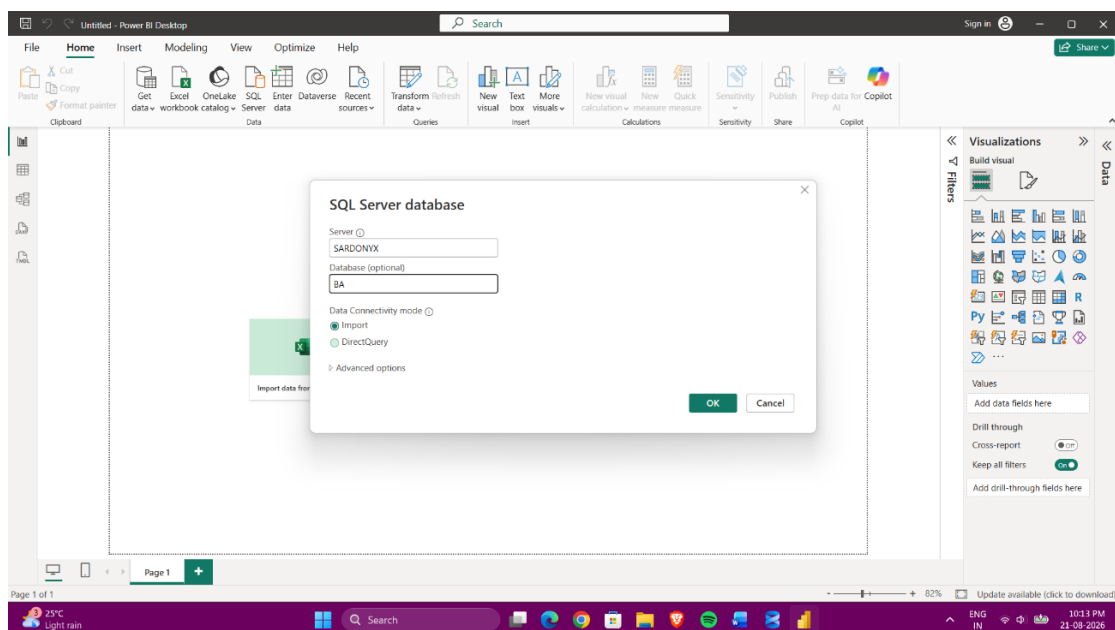
- Obtained Sales and Product Data from CSV/Excel files.
- Connected to MySQL Server and obtained additional data from the database.

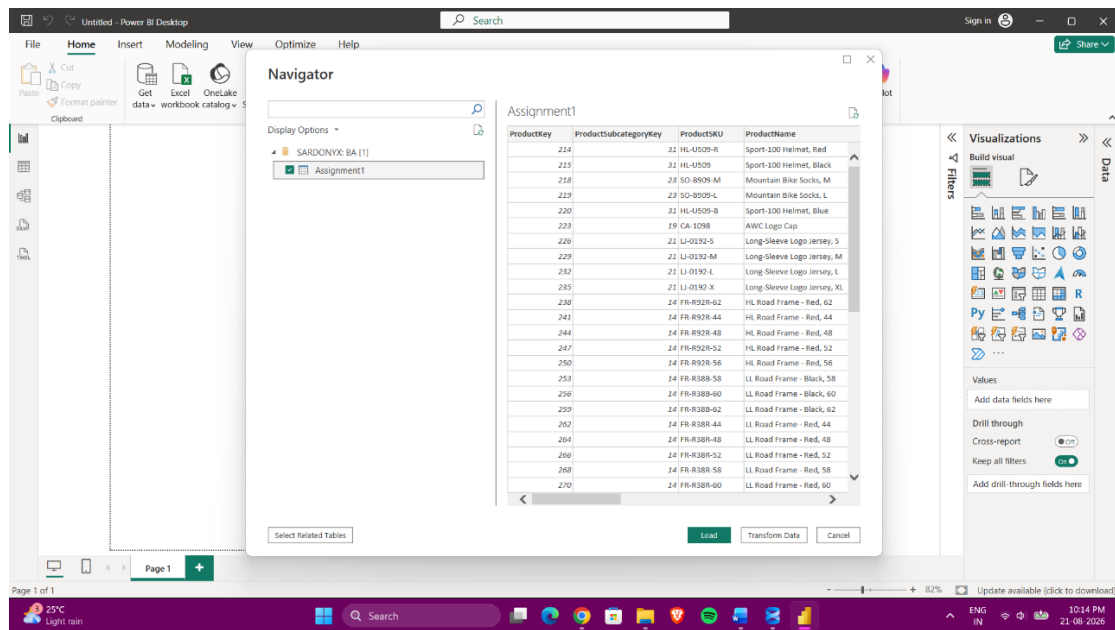
ProductKey	ProductSubcategoryKey	ProductModelKey	ProductName	MakeName	ProductDescription	ProductColor	ProductMaterial	ProductWeight	ProductPrice	ProductCost
1	218	31	MT-1039R	Spoon 100 Mountain Road	Universal 6-speed road bike with lightweight aluminum frame	Red	Al	0	1308399999999999	3400000000000
2	218	31	MT-1039R	Spoon 100 Mountain Road	Universal 6-speed road bike with lightweight aluminum frame	Black	Al	0	1202779999999999	3084415000000
3	218	31	MT-1039R	Spoon 100 Mountain Road	Universal 6-speed road bike with lightweight aluminum frame	White	Al	0	3350333333333333	8000000000000
4	218	23	MT-1039R	Mountain Blue Soda L	Combination of natural and synthetic fibers step 4	White	L	0	2333333333333333	5000000000000
5	218	31	MT-1039R	Spoon 100 Mountain Road	Universal 6-speed road bike with lightweight aluminum frame	Blue	Al	0	1933333333333333	2000000000000
6	223	19	GA-1030	AWG Logo Jersey	Cycling Cap	Black	U	0	3333333333333333	3084415000000
7	223	21	LT-1030	Long-Geeze Logo Jersey	Universal long sleeve AWG logo microfiber cycling jacket	Black	U	0	3175133333333333	4600725000000
8	223	21	LT-1030	Long-Geeze Logo Jersey	Universal long sleeve AWG logo microfiber cycling jacket	Black	M	0	3175133333333333	4600725000000
9	223	21	LT-1030	Long-Geeze Logo Jersey	Universal long sleeve AWG logo microfiber cycling jacket	Black	L	0	3175133333333333	4600725000000
10	223	21	LT-1030	Long-Geeze Logo Jersey	Universal long sleeve AWG logo microfiber cycling jacket	Black	S	0	3175133333333333	4600725000000
11	223	14	FT-1030	FL Road Frame - Road 43	Our lightest and best quality aluminum frame max.	Red	Al	43	7470522222222222	1285455620000
12	241	14	RR-1030	HL Road Frame - Road 43	Our lightest and best quality aluminum frame max.	Red	Al	44	7470522222222222	1285455620000
13	244	14	FT-1030	FL Road Frame - Road 43	Our lightest and best quality aluminum frame max.	Red	Al	43	7470522222222222	1285455620000
14	247	14	RR-1030	HL Road Frame - Road 43	Our lightest and best quality aluminum frame max.	Red	Al	44	7470522222222222	1285455620000
15	250	14	FT-1030	FL Road Frame - Road 43	Our lightest and best quality aluminum frame max.	Red	Al	43	7470522222222222	1285455620000



Step 2: Data Import

- Opened Power BI Desktop.
- Imported the required CSV and Excel datasets using the Get Data option.
- Connected Power BI with the MySQL Server to access the required data.





Step 3: Data Modeling

- Used the available Sales and Product data for analysis.
- Created the required data structure for developing the dashboard.

Step 4: DAX Measures and KPI Creation

- Created new measures using DAX functions.
- Created measures for Average Product Price, Average Product Cost, Average Profit, and Profit Margin.
- Used the measures to create KPI visuals in the dashboard.

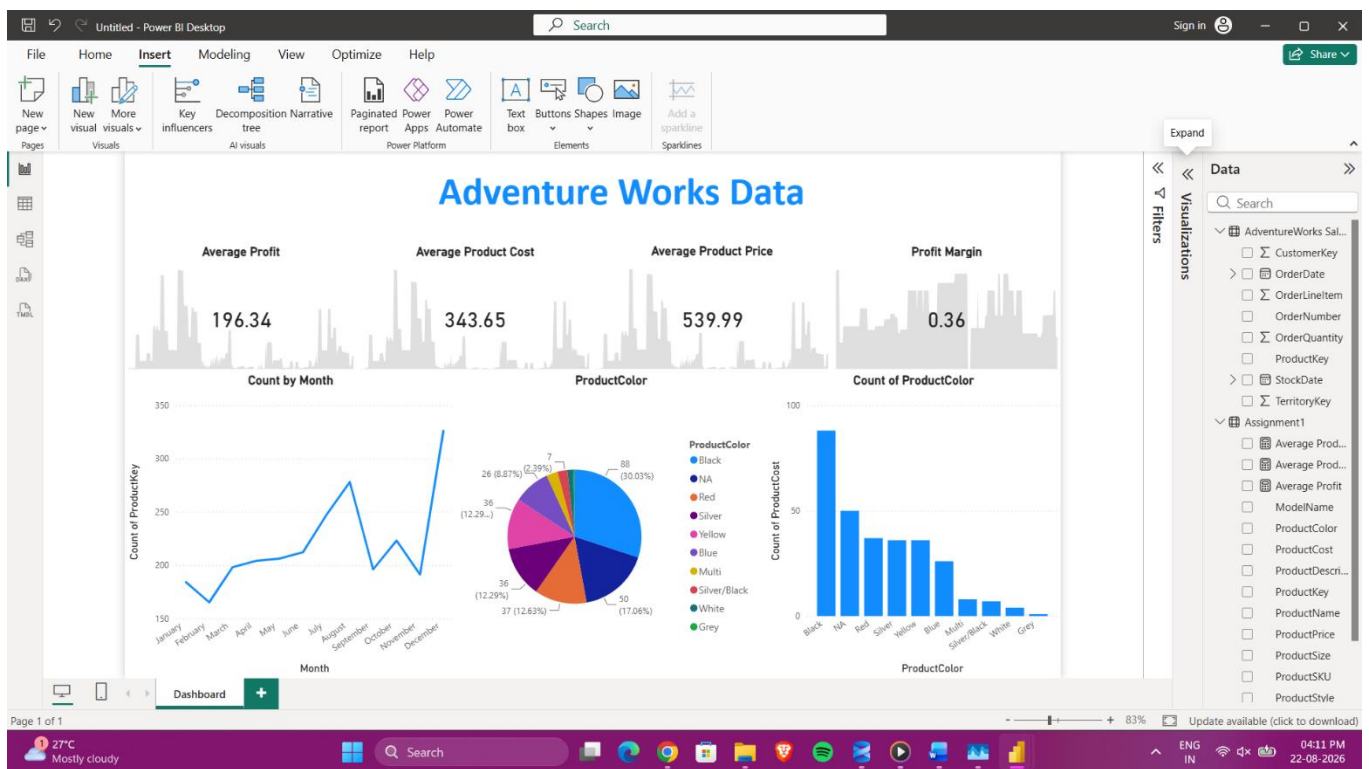
1 Average Product Cost =
2 AVERAGE(Assignment1[ProductCost])

1 Average Profit =
2 AVERAGEX(
3 Assignment1,
4 Assignment1[ProductPrice] - Assignment1[ProductCost]
5)

Step 5: Dashboard Development

- Created KPI visuals for Average Product Price, Average Product Cost, Average Profit, and Profit Margin.
- Developed charts to analyze product data by month and product color.
- Added a line chart to show product count by month.
- Added a pie chart and column chart to analyze products by color.
- Designed an interactive business dashboard for product analysis.

Output



The final Power BI dashboard provides:

- Average Product Price Analysis
- Average Product Cost Analysis
- Average Profit Analysis

- Profit Margin Analysis
- Monthly Product Count Analysis
- Product Color Analysis
- Interactive Dashboard Visuals

The dashboard enables users to analyze product performance and understand pricing, cost, and profitability effectively.

Conclusion

This project successfully combined data from different sources and integrated it into Power BI for effective analysis. The available data was connected, organized, and modeled to generate useful insights. DAX measures were created to calculate key metrics such as average product price, cost, profit, and profit margin. The developed dashboard presents these insights in a simple and interactive manner, helping users understand product performance and support better decision-making.

Learning Outcomes

Through this project, I learned:

- How to import data from CSV and Excel files into Power BI.
- How to connect Power BI with a MySQL Server and access database data.
- How to use existing cleaned data for analysis and dashboard development.
- Creating DAX measures to calculate average product price, product cost, profit, and profit margin.
- Creating and using KPI visuals and different charts for product analysis.
- Designing an interactive Power BI dashboard using different visualizations.
- Analyzing product data and generating meaningful insights from the available data.